# Avery Matherne

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## EDUCATION

**Louisiana State University** Expected May 2028
*M.S., Electrical and Computer Engineering (thesis track), communications and signal processing*

**Louisiana State University** May 2026
*B.S., Computer Engineering, minor in Computer Science*

- Tiger Excellence Non-Resident Award, all four years
- Coursework: Signals and Systems, Digital Communication and Networking, Probability for ECE, Communications in Computing, Communications Engineering Design

## RESEARCH AND ENGINEERING EXPERIENCE

**Graduate Research Assistant, Louisiana State University, Electrical and Computer Engineering** 2026 to present

- Support LSU's side of Intelligent Data Management for Distributed Naval Platforms, a project building decision-making tools that help distributed autonomous maritime systems decide which data matters and how to share it.

**Engineering Intern, Integer Technologies** May 2026 to Aug 2026

- Worked on acoustic characterization and underwater network simulation in support of autonomous maritime systems.

## TECHNICAL PROJECTS

**RAMpage Combat Robot, Senior Design Capstone** Aug 2024 to May 2025
*Lead programmer and systems integration · 1st place, senior design combat robotics competition*

- Wrote the control software for a 30 lb combat robot: MicroPython on a Raspberry Pi Pico W and Python on a Raspberry Pi 4B.
- Designed a split architecture: the Pico W handles the weapon from direct RC PWM input, and the Pi 4B streams video on its own.
- Built pneumatic weapon sequencing with a 17-shot air budget, a kill switch and signal-loss safety stops.

**V2V Collision Warning over OFDM** Jan 2026 to May 2026
*EE 4003 final project, with Jason Phan*

- Built a BPSK-OFDM physical layer in MATLAB, modeled on IEEE 802.11p, and ran it over two ADALM-PLUTO SDRs.
- Implemented the receiver: packet detection, timing recovery, CFO correction, a Costas loop, MMSE equalization and Viterbi decoding.

**16×16 LED Audio Spectrum Visualizer** 2025
*Personal project*

- Built a real-time spectrum visualizer on an ESP32 with a 128-point FFT mapped to 16 log-spaced bands.
- Added an adaptive noise baseline and debounced mode switching between the spectrum and an idle glow.

**Landspace, Land Research and Mapping App** 2025 to present
*Personal project*

- Building a local-first web and desktop app (React, Leaflet, Electron) for mineral title work.
- Imports messy Excel runsheets, maps parcels, and flags problems in the chain of title automatically.

## PROFESSIONAL EXPERIENCE AND LEADERSHIP

**Web Designer and Developer, Freelance** 2026 to present

- Shipped plvinc.com, gracemaealterations.com and becauseofadam.org, each with an editor the owner can use without me.
- Accessible and fast by default: keyboard support, reduced motion, structured data and optimized images.

**Co-Founder, Because of ADAM, 501(c)(3)** 2020 to present

- Lead fundraising and community outreach, from cornhole tournaments to water safety booths.
- Designed and built becauseofadam.org, where every statistic links to its source.

**Land Agent (Contract), Harold J. Anderson Inc. · Lemoine Landman Services · Petro Land Ventures Inc.** 2020 to present

- Research ownership and coordinate permitting for oil, gas and renewable energy projects.
- Build and analyze Excel runsheets to trace title and support project development.

## SKILLS

**Proficient:** Digital communications (BPSK, OFDM, synchronization), Software-defined radio (ADALM-PLUTO), MATLAB, ESP32, Raspberry Pi, Embedded wiring and soldering, Public records and title research, Excel runsheets, Client communication, Project management, Fundraising and outreach

**Familiar:** Python, JavaScript, TypeScript, MicroPython, C and C++, Astro, React, Next.js, Tailwind CSS, Git and GitHub, GitHub Actions, Linux, Playwright, Cloudflare, Vercel, Supabase, Leaflet, OpenCV, Accessibility (WCAG), SEO and structured data, Headless CMS setup, Signal processing (FFT, filtering, AGC), Channel coding (convolutional, Viterbi), Equalization (MMSE)